

**Frankfurt University  
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**Conformal Uncertainty for Operator Deep  
Smoothing of Implied-Volatility Surfaces**

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# Declaration

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# Abstract

Implied volatility surfaces are central to options pricing, hedging, and risk management, yet existing methods for constructing them provide only point predictions with no accompanying measure of uncertainty. This thesis develops a distribution-free uncertainty quantification framework for implied volatility surfaces produced by the Graph Neural Operator (GNO) approach of Gonon, Jacquier, and Wiedemann, which learns a discretisation-invariant map from observed market quotes to smooth surfaces using a single neural network evaluation.

The proposed framework wraps the pre-trained GNO model with a conformal prediction layer that provides finite-sample coverage guarantees without any distributional assumptions. Two economically motivated nonconformity scores are introduced: a spread-normalised price error that measures prediction deviations relative to the bid-ask spread, and a Vega-normalised implied volatility error that weights errors by option sensitivity. A rolling window calibration mechanism adapts the conformal quantiles to non-stationary market conditions by restricting calibration to the most recent market surfaces. Mondrian conformal prediction is investigated as a means of capturing spatial heterogeneity across the strike-expiry domain by computing region-specific quantiles. The resulting uncertainty bands are constructed in both implied volatility and option price spaces, and an iterative projection procedure is applied to enforce no-arbitrage constraints on both band surfaces.

The framework is evaluated on S&P 500 index options data spanning December 2025 to April 2026. Empirical results demonstrate that the conformal bands achieve the nominal 90% coverage target, confirming the validity of the approach on real financial data.

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# Chapter 1

## Introduction

The unprecedented growth of options markets in recent years has created new challenges for financial institutions and traders seeking to manage risk and price derivatives accurately. The Chicago Board Options Exchange (CBOE) reported record-breaking trading volumes in 2025, with 4.6 billion contracts traded across its four exchanges - marking the sixth consecutive year of record growth. Notably, S&P 500 Index (SPX) options alone traded 970.6 million contracts with an average daily volume of 3.9 million contracts [5]. This surge in activity has intensified the demand for robust, real-time methods to construct implied volatility (IV) surfaces from increasingly complex and irregular market data.

Implied volatility transforms the nominal price of an option into a conceptually and numerically convenient metric that reflects the market's expectation of future price fluctuations in the underlying asset. The implied volatility surface represents the collection of these volatilities across different strikes and maturities at a specific point in time, visualized as a three-dimensional surface over the strike-expiry domain. Extracting a smooth, arbitrage-free surface from quoted option prices is a fundamental challenge in quantitative finance, as it directly impacts hedging strategies, risk management decisions, and derivative pricing.

Traditional approaches to implied volatility smoothing rely on parametric surfaces whose parameters are optimized based on the distance to observed prices while adhering to absence-of-arbitrage conditions. The industry-standard Stochastic Volatility Inspired (SVI) model [7] provides a continuous interpolation scheme for each maturity slice, requiring per-instance calibration. More sophisticated parametric methods can be considered proprietary trading secrets and reduce SVI to a benchmark role. However, these parametric approaches share a common limitation: they require continuous execution of numerical optimization routines, which are sensitive to initial conditions, search criteria, and termination criteria. This exposes practitioners to considerable uncertainties during trading hours when markets move rapidly.

Recent advances in machine learning, particularly in neural network architectures, have opened new possibilities for volatility surface construction. The Operator Deep Smoothing approach introduced by Gonon, Jacquier, and Wiedemann [8] represents a significant breakthrough in this domain. Their method employs Graph Neural Operators (GNO) to construct smooth implied volatility surfaces by learning, from irregularly sampled market quotes, a mapping to arbitrage-free surfaces that is invariant to the input grid configuration [8]. Unlike classical parametric methods, this approach performs a single evaluation of a neural network, eliminating the need for instance-by-instance numerical optimization. The GNO architecture naturally handles the irregular spatial arrangement of option quotes and adapts to the dynamically changing configurations of available strikes and maturities.

Despite the remarkable performance of the GNO approach in producing accurate point predictions, a critical gap remains: the model provides no quantification of prediction uncertainty. In financial applications, uncertainty quantification is not merely a theoretical desideratum but a practical necessity. Trading decisions, risk limits, and regulatory capital requirements all depend on reliable uncertainty estimates. Market participants need to understand not only the predicted implied volatility but also the confidence bounds around these predictions, especially in regions of sparse liquidity or during regime shifts.

Traditional approaches to uncertainty quantification in machine learning, such as Bayesian neural networks or ensemble



ble methods, introduce substantial computational overhead and require extensive hyperparameter tuning. Moreover, these methods often rely on distributional assumptions that may not hold in the complex, non-stationary environment of financial markets. An alternative framework that has gained prominence in recent years is conformal prediction, a distribution-free approach that provides finite-sample coverage guarantees under the sole assumption of exchangeability.

Conformal prediction offers an elegant solution to the uncertainty quantification problem. Rather than attempting to model the full predictive distribution, it constructs prediction intervals or regions that are guaranteed to contain the true value with a user-specified probability. The key advantage of conformal prediction lies in its model-agnostic nature: it can be applied as a post-processing wrapper around any point predictor without modifying the underlying model architecture or training procedure [15]. Furthermore, the coverage guarantees are valid for finite samples and do not rely on asymptotic approximations or parametric assumptions.

A key challenge in applying conformal prediction to implied volatility surface forecasting is the design of suitable nonconformity scores. Simple residual-based scores treat all prediction errors equally and therefore ignore the fact that prediction uncertainty varies across different regions of the volatility surface. In practice, pricing errors are influenced by market characteristics such as bid-ask spreads and option Vega, which affect the reliability of observed prices. Consequently, prediction errors should be normalized using market-observable quantities so that the resulting prediction intervals reflect genuine model uncertainty rather than differences in data quality or measurement noise across strikes and maturities.

## 1.1 Motivation and Problem Statement

The central problem addressed in this thesis is the construction of calibrated uncertainty bands for implied volatility surfaces produced by the Graph Neural Operator approach. More specifically, this work develops an efficient uncertainty quantification layer that:

1. Provides finite-sample coverage guarantees without distributional assumptions
2. Accounts for heteroskedastic pricing errors through economically motivated normalization
3. Adapts to temporal regime shifts via rolling calibration windows
4. Produces uncertainty bands in both implied volatility and price spaces, and
5. Satisfies standard no-arbitrage constraints on the resulting band surfaces

This problem is further complicated by the temporal and spatial heterogeneity of options data. Different regions of the strike-expiry domain exhibit varying levels of liquidity, leading to different signal-to-noise ratios. Options far out-of-the-money or with very short maturities tend to have wider bid-ask spreads and less reliable quotes. A robust uncertainty quantification framework must account for this spatial heterogeneity while maintaining valid coverage across the entire surface.

## 1.2 Research Objectives

This thesis pursues the following specific objectives:

1. **Design economically motivated nonconformity scores:** Develop two heteroskedastic scoring mechanisms that normalize prediction errors by (a) bid-ask spreads in price space and (b) Vega-weighted sensitivity in implied volatility space, ensuring uncertainty bands align with trading frictions and option Greeks.
2. **Implement rolling window calibration:** Establish a temporal adaptation mechanism using a sliding window of recent market snapshots to compute conformal quantiles, enabling the framework to track regime changes in market volatility dynamics.

3. **Evaluate spatial heterogeneity via Mondrian conformal prediction:** Investigate whether partitioning the strike-expiry domain into spatial bins and computing region-specific quantiles improves coverage uniformity across areas with varying liquidity.
4. **Construct prediction bands in dual spaces:** Provide practitioners with uncertainty estimates in both implied volatility space (for volatility trading) and price space (for delta hedging and risk management).
5. **Enforce no-arbitrage constraints on prediction bands:** Apply an iterative projection algorithm that enforces butterfly and calendar spread constraints on the constructed uncertainty bands, ensuring that the resulting prediction intervals satisfy the standard absence-of-arbitrage conditions in both the lower and upper bound surfaces.
6. **Validate empirical coverage:** Assess whether the conformal bands achieve the nominal coverage probability  $(1 - \alpha)$  on held-out test data from the S&P 500 options market spanning December 2025 to April 2026.

## 1.3 Contributions

The main contributions of this thesis are:

1. **First application of conformal prediction to neural operator-based volatility surfaces:** To the best of our knowledge, this work represents the first integration of distribution-free uncertainty quantification with operator learning architectures in the context of implied volatility smoothing.
2. **Economically aligned nonconformity scores:** Introduction of two novel scoring mechanisms—spread-normalized price errors and Vega-normalized implied volatility errors—that incorporate market microstructure information into the conformal framework, yielding uncertainty bands that reflect actual trading conditions.
3. **Temporal adaptation via rolling calibration:** Demonstration that maintaining a rolling window of recent calibration data enables the conformal predictor to adapt to non-stationary market conditions while preserving finite-sample coverage guarantees under approximate exchangeability of consecutive market surfaces.
4. **No-arbitrage projection of uncertainty bands:** Development of an iterative projection procedure that enforces butterfly and calendar spread constraints on both the lower and upper prediction band surfaces, ensuring that the uncertainty quantification output is not only statistically valid but also financially consistent.
5. **Empirical validation on real market data:** Comprehensive evaluation on S&P 500 options data spanning December 2025 to April 2026, providing evidence that conformal prediction can deliver reliable, distribution-free uncertainty quantification in a challenging real-world financial application.

## 1.4 Thesis Outline

The remainder of this thesis is structured as follows:

**Chapter 2: Background and Related Work** provides the necessary background on implied volatility surfaces, no-arbitrage constraints, traditional smoothing methods, neural operators, and the theoretical foundations of conformal prediction. We review related work on uncertainty quantification in financial machine learning and position our contribution within this literature.

**Chapter 3: Methodology** presents the technical details of our conformal uncertainty quantification framework. We formally define the problem, introduce the two nonconformity scores (spread-normalized and Vega-normalized), describe the rolling window calibration mechanism, explain Mondrian conformal prediction for spatial heterogeneity, detail the construction of prediction bands in both price and implied volatility spaces, and present the no-arbitrage projection procedure that enforces butterfly and calendar spread constraints on the resulting band surfaces.

**Chapter 4: Implementation** describes the software architecture, data preprocessing pipeline, and integration with the pre-trained GNO model from Gonon et al.

**Chapter 5: Experiments and Results** reports the empirical evaluation on S&P 500 options data. We present coverage validation results, visualizations of uncertainty bands, comparisons of different nonconformity scores, and ablation studies on window size and spatial binning.

**Chapter 6: Discussion** interprets the results, examines the practical implications for risk management and trading, discusses the limitations of the approach (particularly regarding the exchangeability assumption), and considers the financial validity of the uncertainty bands.

**Chapter 7: Conclusion** summarizes the contributions, reflects on the broader significance of distribution-free uncertainty quantification for financial machine learning, and outlines directions for future work, including extensions to online conformal prediction, adaptive spatial binning, and integration with richer option pricing models.

## Chapter 2

# Background and Related Work

This chapter establishes the technical foundations on which the thesis builds. We begin by introducing the financial setting of implied volatility surfaces and the no-arbitrage conditions they must satisfy. We then survey classical and machine learning-based approaches to implied volatility smoothing, situating the Graph Neural Operator method of Gonon, Jacquier, and Wiedemann [8] within this landscape. Finally, we review the theoretical framework of conformal prediction and discuss related work on uncertainty quantification in financial machine learning.

### 2.1 Implied Volatility and the Options Market

#### 2.1.1 European Options and the Black-Scholes Model

We consider a market of European options written on an underlying asset observed at a reference time  $T_0$ . Each option is identified by its expiry  $T \in (T_0, \infty)$  and its strike  $K \in (0, \infty)$ . The time-to-expiry is denoted  $\tau = T - T_0$  and the log-moneyness is defined as  $k = \log(K/F_{T_0,T})$ , where  $F_{T_0,T}$  denotes the time- $T$  forward price of the underlying asset.

The Black-Scholes model [4] provides a closed-form expression for the price of a European call option with time-to-expiry  $\tau$  and log-moneyness  $k$  as a function of the volatility parameter  $v \in (0, \infty)$ :

$$\text{BS}(\tau, k, v) = \Phi(d_1(\tau, k, v)) - e^k \Phi(d_2(\tau, k, v)), \quad (2.1)$$

where  $\Phi$  denotes the cumulative distribution function of the standard normal distribution and  $d_1, d_2$  are the standard Black-Scholes terms involving  $\tau, k$ , and  $v$ . The sensitivity of the option price with respect to the volatility parameter, known as Vega, plays a central role in this thesis and is given by:

$$\mathcal{V}(\tau, k, v) = \varphi(d_1(\tau, k, v))\sqrt{\tau}, \quad (2.2)$$

where  $\varphi$  denotes the standard normal density [4].

#### 2.1.2 Implied Volatility

While the Black-Scholes model is no longer considered an adequate model of market dynamics, its mathematical tractability makes it a universal quoting convention. Given a market-observed call option price  $C(T, K)$ , its implied volatility  $v(\tau, k)$  is defined as the unique solution to:

$$C(T, K) = F_{T_0,T} \cdot \text{BS}(\tau, k, v(\tau, k)). \quad (2.3)$$

Implied volatility thus transforms a nominal option price into a dimensionless, scale-invariant metric that allows direct comparison of options across different strikes, maturities, underlyings, and interest rate environments. It reflects the market's collective expectation of future price variability in the underlying asset over the life of the option.

### 2.1.3 The Implied Volatility Surface

At any given instant, an options market offers contracts across a range of strikes and maturities. Inverting the Black–Scholes formula for each quoted option price yields a corresponding implied volatility. The collection of these values, viewed as a function of time-to-expiry  $\tau$  and log-moneyness  $k$ , defines the implied volatility surface. This surface provides a three-dimensional snapshot of how the market prices uncertainty across strikes and maturities.

The implied volatility surface exhibits two well-documented structural features. Across strikes at a fixed maturity, implied volatility typically forms a skewed smile, reflecting the market’s asymmetric assessment of upside and downside risks. Across maturities at a fixed strike or log-moneyness, it displays a term structure that may rise or fall depending on prevailing market conditions. Together, these features encode the market-implied risk-neutral marginal distributions of the underlying asset across maturities, making the surface a central input for derivative pricing, hedging, and portfolio risk measurement.

Because options trade only at a discrete and irregularly spaced set of strikes and maturities, and because this set changes as contracts expire and new ones are listed, the implied volatility surface is not directly observable as a continuous object. It must instead be reconstructed from noisy market quotes through interpolation, extrapolation, and smoothing. This estimation problem, known as implied volatility smoothing, is the central challenge addressed in this thesis.

### 2.1.4 No-Arbitrage Conditions

Not every smooth surface defined over the strike-expiry domain corresponds to a valid market. A surface is said to be free of static arbitrage if it does not permit costless trading strategies constructed from fixed positions in options. Gatheral and Jacquier [7] show that this requirement reduces to two independent conditions, as formalised in Theorem 2.1 of Gonon et al. [8]:

- **No-calendar-arbitrage:** Option prices must increase consistently with maturity. A longer-dated option must never be systematically cheaper than a shorter-dated one at the same strike, as this would permit riskless profit through calendar spread positions.
- **No-butterfly-arbitrage:** The implied volatility smile at each fixed maturity must be sufficiently convex in strike, ensuring that the risk-neutral density of the underlying asset implied by the surface remains non-negative at all strikes.

Both conditions play a direct role in this thesis, as the no-arbitrage projection step of the proposed framework enforces them on the constructed uncertainty bands.

## 2.2 Implied Volatility Smoothing

### 2.2.1 Parametric Approaches

The task of implied volatility smoothing refers to fitting a smooth, arbitrage-free surface  $\hat{v}$  to a finite collection  $\mathbf{v} = \{v(\tau_i, k_l)\}_{l=1}^P$  of observed implied volatilities. Naive interpolation schemes such as cubic splines are ill-suited for this purpose, as they yield arbitrage violations and do not generalise across maturities.

The industry-standard approach is the Stochastic Volatility Inspired (SVI) model [7], which parameterises each maturity slice independently using a five-parameter functional form that captures the characteristic skew and smile of implied volatility. Its extension to full surfaces via the Surface SVI (SSVI) parameterisation [7] yields a globally consistent surface with closed-form no-arbitrage conditions. SVI and its variants have been widely adopted by practitioners.

Despite their practical success, parametric models share a fundamental limitation: they require the execution of a numerical optimisation routine for every new set of market quotes, with no guarantee of convergence to a consistent solution.

## 2.2.2 Neural Network-Based Approaches

The limitations of parametric calibration have motivated a growing body of research on neural network-based implied volatility smoothing. Akerer et al. [1] introduced deep neural network architectures combined with no-arbitrage soft constraints for volatility surface construction.

A related line of work applies variational autoencoders to volatility smoothing [3], exploiting the fixed-grid structure of FX options markets where strikes are quoted at standard deltas. This specificity limits direct application to equity index options such as SPX, where strikes and maturities change continuously as options expire and new ones are listed.

A common limitation of classical neural network architectures in this setting is the requirement for fixed-size inputs. Options data is inherently irregular: at each point in time, a different collection of strikes and maturities is available. This necessitates extensive pre-processing to map market data onto a fixed representation before applying standard feedforward or convolutional networks, introducing information loss and additional engineering complexity.

## 2.2.3 Operator Deep Smoothing

Gonon, Jacquier, and Wiedemann [8] address the irregular structure of options data using the framework of neural operators. In their approach [8], the implied volatility observations available at a given time are interpreted as discrete samples from an underlying smooth function. This function can be written as

$$v : D \rightarrow \mathbb{R},$$

where the domain

$$D = (\rho_{\min}, \rho_{\max}) \times (z_{\min}, z_{\max})$$

is defined using the transformed coordinates [8]

$$\rho = \sqrt{\tau}, \quad z = \frac{k}{\rho}.$$

These coordinates are used because they make the option data more evenly distributed across the input domain.

Under this interpretation, the smoothing problem is no longer treated as a standard interpolation problem. Instead, it is formulated as learning an operator

$$F : \mathcal{A} \rightarrow \mathcal{U}$$

between function spaces. This operator takes the observed, discretised implied volatility data as input and produces a smooth implied volatility surface as output.

The authors adapt the Graph Neural Operator (GNO) architecture [11] to this setting. A GNO represents market observations as point-wise samples of an underlying function, rather than as values on a fixed grid [8]. This is important for options data because market quotes are usually observed at irregular and changing strike-maturity locations. A key modification in their architecture is the removal of the pointwise local linear transformation in the first layer [8]. This makes the first operation purely non-local, allowing the model to process different sets of input locations in a consistent way. As a result, the architecture is discretisation-invariant and can be applied directly to changing market quote locations without requiring the data to be mapped onto a fixed grid beforehand.

The trained operator is applied in a single forward pass to produce the full smoothed surface, eliminating the instance-by-instance optimisation required by parametric approaches. The model is trained on nine years of intraday S&P 500 options data (2012–2020) and demonstrates substantially lower mean absolute percentage error (MAPE) compared to the SVI benchmark on the held-out 2021 test year, while producing surfaces that are empirically free of arbitrage.

A critical limitation of the GNO approach, explicitly noted by the authors, is that it provides only point predictions: the model outputs a single smoothed surface without any accompanying measure of prediction uncertainty. This gap motivates the present thesis.

## 2.3 Conformal Prediction

### 2.3.1 Foundations

Conformal prediction is a framework for constructing distribution-free prediction sets with finite-sample coverage guarantees, originally developed by Vovk, Gammerman, and Shafer and subsequently surveyed by Shafer and Vovk [15]. Unlike Bayesian credible intervals or asymptotic confidence intervals, conformal prediction makes no parametric assumptions about the data-generating process. Its main assumption is that the calibration and test data are exchangeable, meaning that they are generated in the same way and that their ordering does not carry additional information.

Given a fitted point predictor and a calibration dataset

$$\{(X_i, Y_i)\}_{i=1}^n,$$

a nonconformity score  $A(X_i, Y_i)$  is computed for each calibration point. This score measures how unusual the observation is with respect to the fitted predictor. The empirical  $(1 - \alpha)$ -quantile of the calibration scores, denoted by  $\hat{q}_{1-\alpha}$ , is then used to construct a prediction set for a new test point  $X_{n+1}$ :

$$\mathcal{C}(X_{n+1}) = \{y : A(X_{n+1}, y) \leq \hat{q}_{1-\alpha}\}. \quad (2.4)$$

Under exchangeability, this set satisfies the marginal coverage guarantee

$$\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1})) \geq 1 - \alpha. \quad (2.5)$$

This guarantee means that, over repeated test observations, the prediction set contains the true response with probability at least  $1 - \alpha$ . It is non-asymptotic and holds for any sample size and any underlying predictor.

### 2.3.2 Inductive Conformal Prediction

The computationally practical version of conformal prediction is the inductive (or split) conformal predictor [12, 10]. It uses a held-out calibration set that is separate from the training data, computing nonconformity scores only once rather than retraining the predictor for each test point. This separation makes inductive conformal prediction directly applicable as a post-hoc wrapper around any pre-trained model, which is the approach adopted in this thesis.

### 2.3.3 Nonconformity Scores

The choice of nonconformity score determines the shape and adaptivity of the resulting prediction sets. The simplest choice for regression is the absolute residual  $A(x, y) = |\hat{f}(x) - y|$ , which yields symmetric prediction intervals of constant width. More adaptive scores normalise the residual by an estimate of local uncertainty, producing intervals that are wider in regions where the model is less reliable. Conformalized Quantile Regression (CQR) [13] uses quantile regression estimates to produce locally adaptive prediction intervals with valid marginal coverage. The design of economically motivated nonconformity scores tailored to implied volatility is one of the central contributions of this thesis.

### 2.3.4 Mondrian Conformal Prediction

Mondrian conformal prediction [15] extends the basic framework to conditional coverage by partitioning the input space into disjoint bins and computing separate conformal quantiles within each bin. If the bins are chosen to reflect regions of homogeneous uncertainty, Mondrian conformal prediction provides approximate conditional coverage guarantees - that is, coverage that holds not only marginally but also within each spatial region. This is particularly relevant for implied volatility surfaces, where uncertainty varies systematically across the strike-expiry domain.

### 2.3.5 Conformal Prediction for Time Series

A key challenge in applying conformal prediction to financial data is the violation of the exchangeability assumption. Financial time series exhibit serial dependence and non-stationarity: the distribution of market data in one regime may differ substantially from that in another. Several approaches have been proposed to address this in the time series setting. Rolling or sliding window calibration [2] restricts the calibration set to the most recent observations, ensuring that the quantile reflects current market conditions rather than historical ones. Under the assumption that consecutive observations are approximately exchangeable over short time horizons - which is more plausible than global exchangeability - rolling window calibration retains approximate coverage guarantees while adapting to regime changes. This approach is adopted in this thesis.

## 2.4 Uncertainty Quantification in Financial Machine Learning

Uncertainty quantification in financial machine learning has received growing attention, driven by regulatory requirements and the practical need for reliable confidence measures in trading and risk management. Bayesian neural networks [6] and deep ensembles [9] have been applied to financial forecasting, but both introduce substantial computational overhead and rely on distributional assumptions that may not hold in non-stationary market environments. Conformal prediction offers a model-agnostic alternative that avoids these limitations.

In the context of options pricing and volatility modelling, uncertainty quantification remains relatively under-explored. Wiese et al. [16] and Ruf and Wang [14] have studied neural network approaches to options pricing, but without uncertainty estimates. To the best of our knowledge, no prior work has applied conformal prediction to neural operator-based implied volatility smoothing. This thesis addresses that gap.

## 2.5 Summary

This chapter has established the financial and statistical foundations of the thesis. The implied volatility surface is a central object in options markets, subject to strict no-arbitrage constraints. Classical parametric methods calibrate this surface instance-by-instance, while the operator deep smoothing approach of Gonon et al. [8] replaces this with a single neural network evaluation, at the cost of providing no uncertainty estimates. Conformal prediction provides a principled, distribution-free framework for constructing prediction sets with finite-sample coverage guarantees, which can be applied as a post-hoc wrapper around any point predictor. The rolling window and Mondrian extensions adapt this framework to the temporal and spatial heterogeneity of options data. The following chapter presents the methodology that combines these elements into a coherent uncertainty quantification framework for implied volatility surfaces.



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